Package hvqrurl
Creating a QR-code of an URL in the margin

ver 0.02

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This package allows to draw an URL as a QR code into the margin of a one- or twosided document. The following packages are loaded by default: qrcode, marginnote, url, xcolor, and xkeyval.

1 The macros

```
\hvqrset{key=value, ...}
\hvqrurl [key=value, …] {URL}
\hvqrurl* [key=value, …] {URL}
```

With the default macro \hvqrurl the URL is printed as as QR code into the margin and in the the text as usual with the macro \url, for example https://mirror.ctan.org/pkg/hvqrurl. With hyperref you'll get the the same color for the QR code as for the URL link and, of course, is also a link. This example shows the default setting for a QR code.

With \hvqrset one can set the optional arguments globally. For example if one do not want all QR codes not as a link when using hyperref:

```
\hvqrset{qrlink=nolink}
```

2 Optional arguments

2.1 No link with hyperref

```
With qrlink=nolink the QR code is no link: \hvqrurl[qrlink=nolink]{https://mirror.ctan.org/pkg/
    hvqrurl}.
The default setting is qrlink=link.
```

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With qrlink=nolink the QR code is no link: https://mirror.ctan.org/pkg/hvqrurl. The default setting is qrlink=link . Without using hyperref this optional argument has no meaning.

2.2 Color of the QR code

Without using hyperref the default color is black. It can be changed by the optional argument qrcolor. The package xcolor is loaded by default, the reason why an extended color definition is possible. For this example we used

```
The package xcolor\hvqrurl*[qrcolor=red!40!white]{http://mirror.ctan.org/pkg/xcolor}
 is loaded by default, ...
```

2.3 Vertical position of the QR code

By default the baseline of the QR code is nearly at the same height as the baseline of the textline. However, when changing the size of the QR code it may be nessesary to move up or down the QR code. The default value of qradjust is -1.5\normalbaselineskip. Setting the value to 0pt the QR code is moved down which is the default setting without a vertical adjustment.

```
The default value of qradjust is \verb|-1.5\normalbaselineskip|. Setting the value
to 0pt the QR code \hvqrurl*[qradjust=0pt]{http://ctan.org/} is moved down which is the default
setting without a vertical adjustment.
```

2.4 Size of the QR code

By default the QR code is a square with height and width of 1cm. it can be changed by setting qrheight to another value, for example to 2cm: This is an exetremely long url where it may make sense to use a larger QR code.

```
it can be changed by setting qrheight to another value, for example to 2cm:
\hvqrurl*[qrheight=2cm]{https://identity.fu-berlin.de/idp-fub/profile/SAML2/Redirect/SSO;
    jsessionid=71C984647E3B8F2E716CA067CB13387E?execution=e1s1}
This is an exetremely long url where it may make sense to use a larger QR code.
```

2.5 QR code level

The QR code specification includes four levels of encoding: Low (L) (https://www.tug.org/), Medium (M), Quality (Q), and High (H) (https://www.tug.org/), in increasing order of error-correction capability. In general, for a given text a higher error-correction level requires more bits of information in the QR code.

```
The QR code specification includes four levels of encoding:
Low (L) (\hvqrurl{https://www.tug.org/}), Medium (M), Quality (Q),
and High (H) (\hvqrurl[qradjust=0pt,qrlevel=H]{https://www.tug.org/}),
in increasing order of error-correction capability.
```

The first QR code has the default level M and the last one the level H. In general the user has not to set this keyword it will be controlled internally by the package.

2.6 Inner or outer margin for the QR code

By default the QR code is set in the right (oneside document) or outer margin (twoside) of the document. This one (http://hvoss.org) is at the default position. With the optional argument `qrreverse` it can be placed in the left or inner margin. This one (http://latex.texnik.de) is in the other margin.

```
This one (\hvqrurl{http://hvoss.org}) is at the default position.
With the optional argument \Lkeyword{qrreverse} it can be placed in the left or inner margin.
This one (\hvqrurl[qrreverse]{http://latex.texnik.de}) is in the other margin.
```

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```
%% $Id: hvqrurl.sty 1123 2019-12-01 09:16:34Z herbert $
%%
%% This file is distributed under the terms of the LaTeX Project Public
%% License from CTAN archives in directory  macros/latex/base/lppl.txt.
%% Either version 1.3 or, at your option, any later version.
%%
% Copyright 2019,21 Herbert Voss hvoss@tug.org
%%
\ProvidesPackage{hvqrurl}[%
  2021/05/18 v.0.02 (Herbert Voss) Supports qr images into the margin (hv)]
%
\NeedsTeXFormat{LaTeX2e}
\RequirePackage{url}
\RequirePackage{xcolor}
\RequirePackage{marginnote}
\RequirePackage{qrcode}
\RequirePackage{xkeyval}

\newlength\qr@url@qrheight
\newlength\qr@url@qradjust

\define@key{hvqr}{qrheight}[1cm]{\setlength\qr@url@qrheight{#1}}
\define@key{hvqr}{qradjust}[-1.5\normalbaselineskip]{\setlength\qr@url@qradjust{#1}}
\define@key{hvqr}{qrcolor}[black]{\colorlet{qr@url@qrcolor}{#1}}
\define@key{hvqr}{qrlevel}[M]{\def\qr@url@qrlevel{#1}}
\define@key{hvqr}{qrlink}[link]{\def\qr@url@qrlink{#1}}
\define@boolkey{hvqr}[hv@]{qrreverse}[true]{}%     use of the hvOBox contents

\setkeys{hvqr}{qrheight,qrcolor=black,qradjust,qrlevel,qrlink=link,qrreverse=false}% the default setting

\newcommand*\hvqrset[1]{\setkeys{hvqr}{#1}}

\def\hvqrurl{\@ifnextchar*{\@tempswafalse\hvqr@url}{\@tempswatrue\hvqr@@url}}

\def\hvqr@url*{\hvqr@@url}

\newcommand*\hvqr@@url[2][]{%
  \begingroup
  \providecommand*\qr@blank{0}%
  \ifx\relax#1\relax \else\hvqrset{#1}\fi
  \expandafter\qrset\expandafter{\qr@url@qrlink,height=\qr@url@qrheight,level=\qr@url@qrlevel}%
  \ifhv@qrreverse\reversemarginpar\fi
  \if@tempswa\url{#2}\fi
  \edef\reserved@a{\noexpand\marginnote{\noexpand\color{qr@url@qrcolor}\noexpand\qrcode{%
     \detokenize\expandafter{#2}}}[\qr@url@qradjust]}%
  \reserved@a
  \endgroup
}

\endinput
```